Qualifying Spatial Relations using Jev:

What role could artificial system-one intelligence play in GeoAI?

Disposition for a master thesis by Jakob Roland
Supervised by Prof. Dr. Krzysztof Janowicz
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Introduction

Since the release of the first chatbot powered by a large language model (LLM) in late 2022, it has become clear that artificial intelligence (AI) has left the realm of science fiction to enter our daily life. Since then almost all spheres of society have been impacted in some way by AI, Spatial Data Science being no exception. There have been many works testing LLMs for a variety of spatial tasks [EXAMPLES] and some discussing their caveats and ethical implications (e.g. [1, 3]). This thesis hopes to contribute to the scientific discourse surrounding GeoAI and sustainability in testing the spatial understanding of the system-one intelligence model Jev [cite system one], which was released by the company Typesafe AI in September 2026 and promises efficient data processing at scale. To this end, the experiment on spatial relation qualification done by Yuhan Ji (et al, 2025 [2]) is reproduced using Jev instead of ChatGPT-4. Compared to ChatGPT-4 Jev is quicker, cheaper, and responds in a predetermined format, which facilitates downstream processing of the results. The question remains if the accuracy of Jev's answers is comparable to GPT-4, which this thesis will attempt to find out.

Research Goals

GeoAI has been tested for many applications in the field of spatial data science [EXAMPLES], one of them being spatial relation qualification [2]. Such exploration helps us understand how LLMs represent and analyze spatial information and how (agentic) AI can aid in geoprocessing tasks (cite GeoAutomatom). At the same time there are ethical concerns with GeoAI [1, 3], a major example being sustainability. Geodata is usually high in volume and temporal resolution, thus requiring many processing steps and, especially if LLMs are involved, substantial compute. Typesafe AI's Jev is set up to process high volumes of data efficiently, quickly, and reliably, which makes it seem suitable for sustainable GeoAI.

This thesis will put this assumption to the test, reproducing the experiment

of spatial relation qualification made by Yuhan Ji (et al, 2025 [2]) using Jev, enabling a direct comparison between ChatGPT-4 and Jev and highlighting advantages and disadvantages of both models. Doing so the thesis hopes to answer the following research questions:

- How does Jev fare with spatial relation qualification?
- In what ways could system-one intelligence models benefit GeoAI, especially considering aspects of sustainability?
- Do proven prompt engineering methods like few-shot prompting and Chain-of-Thought reasoning work for system-one intelligence models?

Methodology

To ensure a good comparison this thesis will stay as true as possible to the original experiment in [2], using the same data and setup. The prompts will have to be adapted to the unique requirements of Jev. The experiment will utilize Typesafe AI's API and Python SDK to utilize Jev. All code and results will be made available together with the thesis on Github. A short review of related works will accompany the experiment.

Expected Results and Challenges

Initial experimentation proved Jev's speed and (cost-)efficiency, while it could not beat the benchmark set by ChatGPT-4 in terms of accuracy. Still, the confidence score, which Jev assigns to its answers, seems reliable, and some interesting patterns occurred in the inital results, which merit further investigation.

The main challenge this thesis will face is a problem common with all LLMs: Their opacity. Jev's answers especially lack any reasoning and cannot be reproduced reliably. In addition, Jev is a proprietary model, with very little official information released publicly.